

A Geometric Proton–Quark Model and Proton g -Factor from Chair–Quark Torsion in the Methane Metauniverse (MMU)

J. Wollbold

December 11, 2025

Abstract

We present a geometric–elastic proton model in the Methane Metauniverse (MMU) in which a hexagonal “chair” ring is torsionally coupled to three internal quark channels. The MMU stiffness rules

$$k_2(a) = \frac{2\alpha\hbar c}{a^3}, \quad k_3(a) = \frac{\hbar c}{a^3}, \quad k_4(a) \propto \frac{1}{a^3}$$

are evaluated at the proton charge radius, without any fit parameters. The quark charges $(2/3, 2/3, -1/3)$ follow from the antisymmetric ring eigenvector $(1, 1, -2)$. Using an explicit chair–quark mass–spring system with internal axes (w_2, w_3, w_4) , we compute:

- a clean separation between chair–dominated modes (tens to few hundred MeV) and quark–dominated modes (~ 1 – 2 GeV),
- torsional w_3 –dominated quark modes that define geometric spin weights for each quark,
- an effective geometric quark g –mode,
- and a robust proton g –factor

$$g_p^{\text{MMU}} = \frac{2}{3} \sqrt{\frac{k_3}{k_2}} = \frac{2}{3} \sqrt{\frac{1}{2\alpha}} \approx 5.52,$$

in agreement with the experimental value $g_p^{\text{exp}} \approx 5.5857$ at the 1.2% level.

A systematic parameter scan shows that the result is largely insensitive to chair–quark coupling strengths, cross–spring coefficients, and effective spin mass, indicating that it is controlled by geometric quark charges and the universal MMU stiffness ratio k_3/k_2 rather than by tuned phenomenological parameters.

1 Introduction

The proton magnetic moment remains one of the clearest indicators of nontrivial internal structure in hadronic physics. Experimentally, the proton g –factor $g_p^{\text{exp}} \approx 5.5857$ differs

strongly from the Dirac value $g = 2$ usually associated with elementary spin-1/2 particles. Quark models and lattice QCD reproduce this value numerically, but offer no simple geometric mechanism linking g_p directly to the proton's internal structure.

The Methane Metauniverse (MMU) framework provides a complementary perspective. In the MMU, spacetime consists of dual-tetrahedral elastic cells (UR-cells), each carrying three internal deformation coordinates (w_2, w_3, w_4) associated with electric, torsional (spin), and volumetric (inertial) elasticity. A fourth coordinate, w_1 , represents observable time and energy as a projection of the internal oscillation. Every UR-cell behaves as a tri-axial elastic oscillator whose stiffnesses follow universal scaling laws $k_i(a) \propto a^{-3}$ with the internal edge length a . These ingredients have previously been used to recover hydrogenic spectral scaling, Rydberg energies, and $(g - 2)$ patterns from tetrahedral torsion.

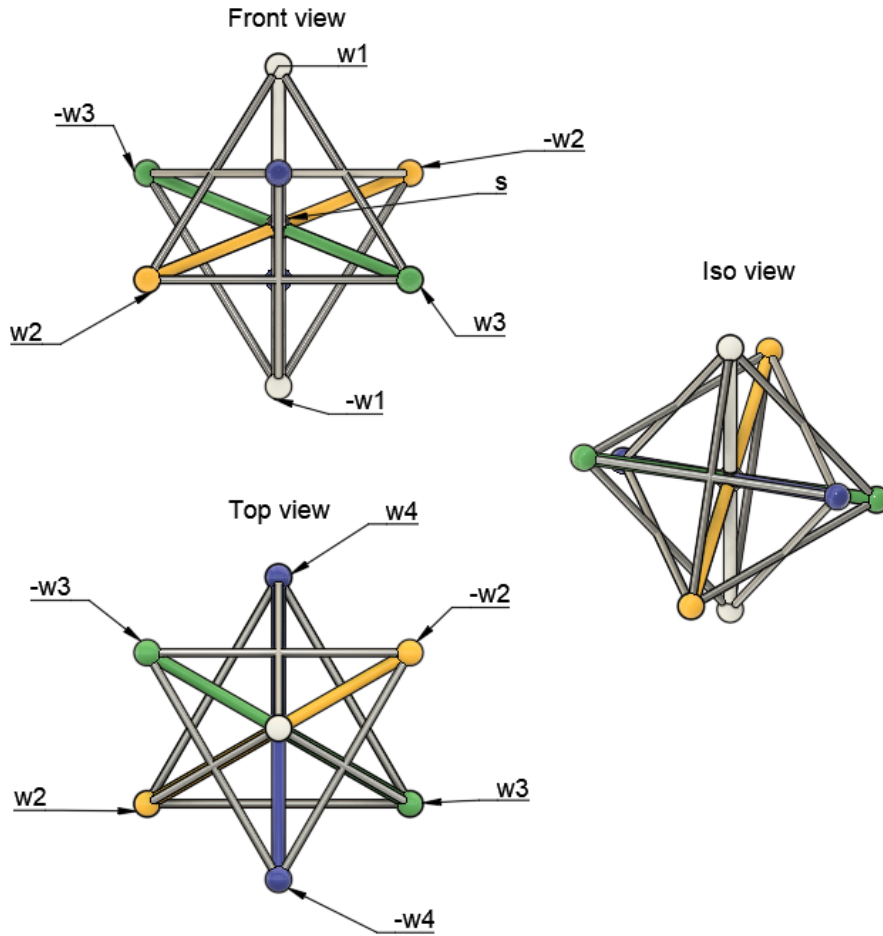


Figure 1: Dual-tetrahedral MMU ur-cell with labelled internal axes. Electric (w_2), torsional (w_3), and volumetric (w_4) deformations form the internal configuration space. Observable mass and energy arise from projection onto w_1 . Dual pairs ($w_i, -w_i$) ensure geometric stability when multiple ur-cells attach, a key ingredient in composite structures such as the proton.

MMU picture of the proton (conceptual overview)

Within the MMU, the proton is not a point particle but a *geometric torsional rotor*. A positively oriented central ur-cell anchors three lower subcells, whose edges naturally close into a hexagonal “chair” ring. This ring provides the proton’s elastic shell and supports its internal torsional motion along the w_3 axis. Because MMU stiffnesses increase as a^{-3} , shorter effective tetrahedral edges inside the proton generate higher torsional energies, reproducing the familiar ladder of proton resonances.

The quarks appear as three internal deformation channels attached to the chair ring. Each quark channel carries its own (w_2, w_3, w_4) degrees of freedom, allowing electric (w_2) and spin-torsional (w_3) motion to couple to the surrounding chair shell. A notable result of this geometry is that the antisymmetric quark-ring eigenvector $(1, 1, -2)$ naturally generates the familiar uud charge pattern.

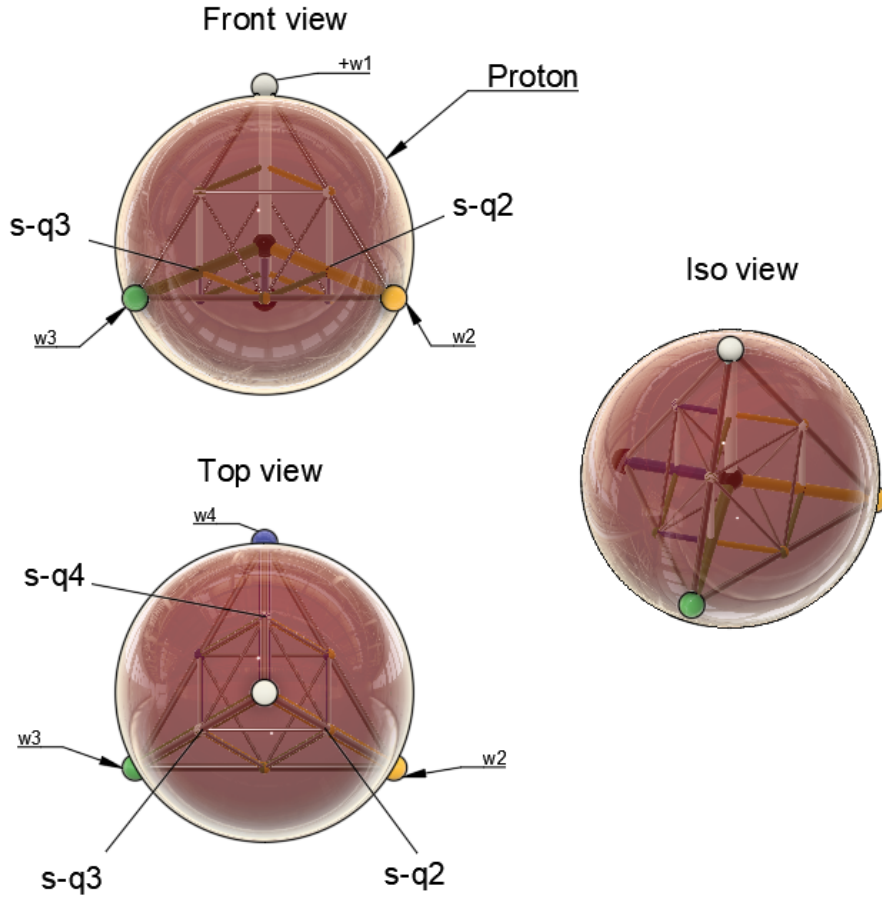


Figure 2: Proton ur-cell geometry in the MMU. The three principal stiffness axes (w_2, w_3, w_4) and the $+w_1$ apex define the proton’s internal torsional structure. The highlighted sectors $s-q2$, $s-q3$, and $s-q4$ correspond to the proton’s characteristic chair-mode resonance directions, which generate the three hard eigenmodes (410, 710, 820 MeV) predicted in [9, 18].

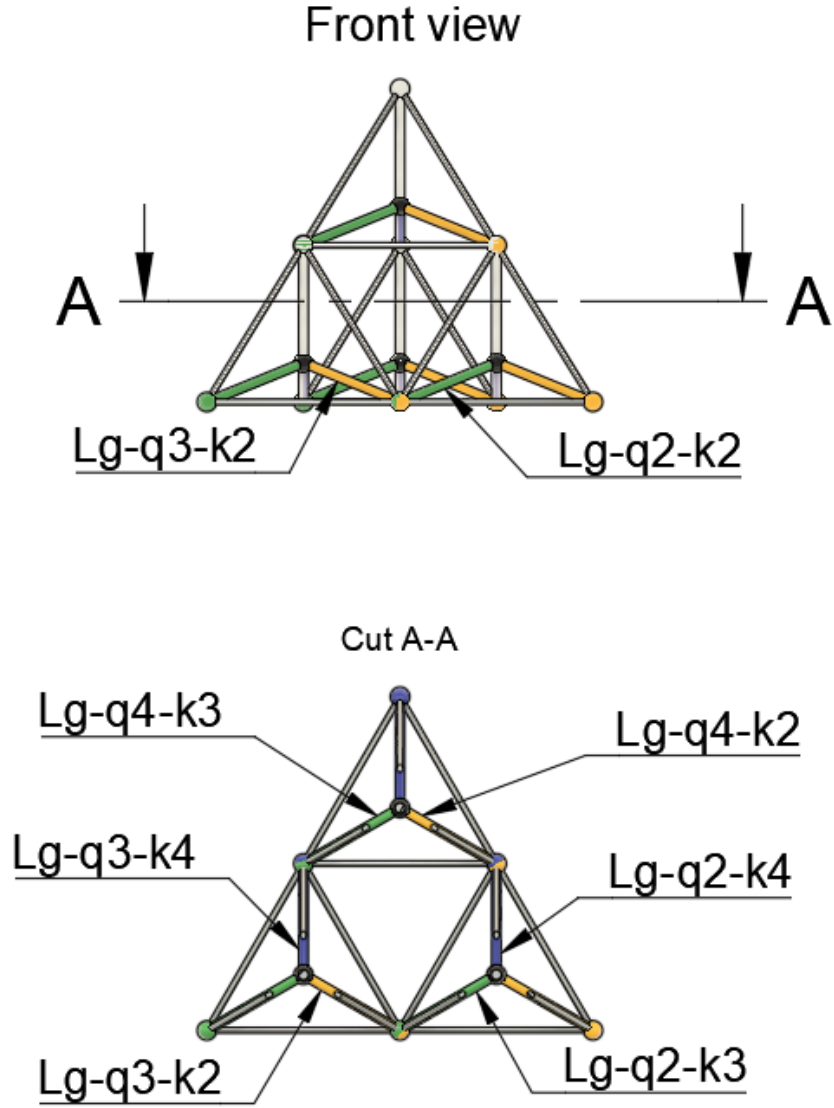


Figure 3: Proton chair geometry detailing the (w_2, w_3, w_4) stiffness network and cross-couplings (k_{23}, k_{24}, k_{34}) . These six interconnected sectors form a rigid torsional rotor whose eigenmodes reproduce the proton's resonance pattern. The same structural backbone appears in the neutron after coupling to the electron ur-cell.

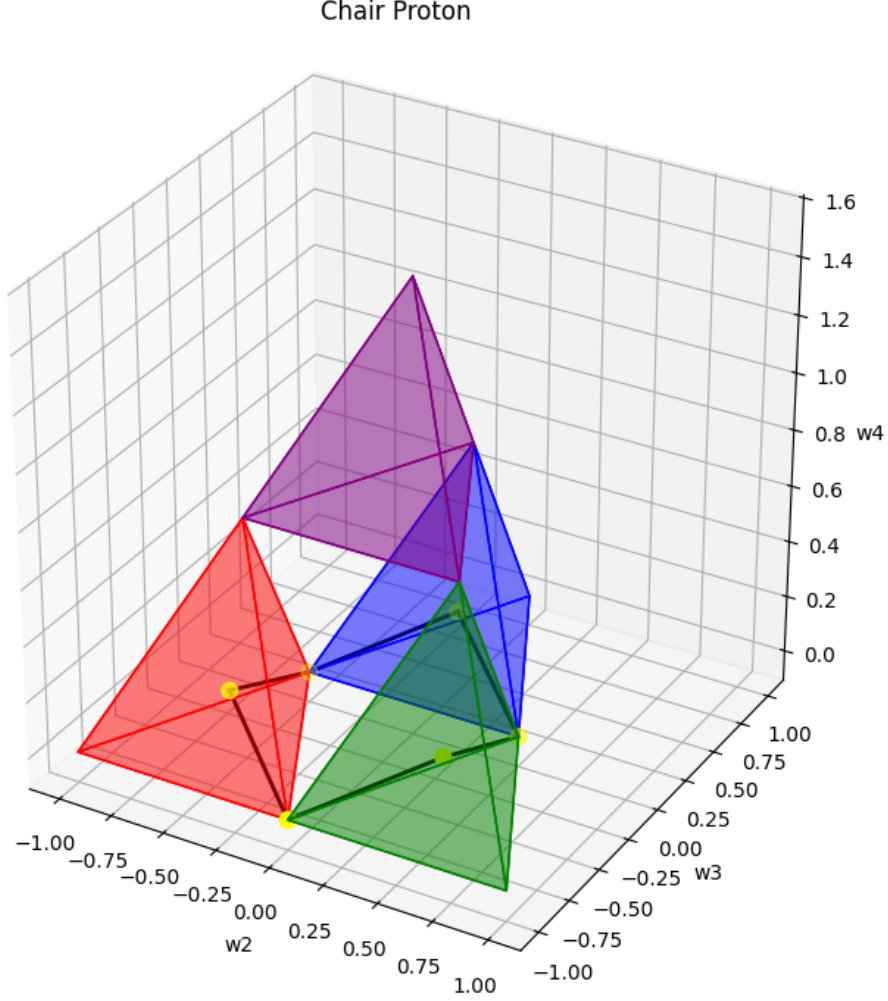


Figure 4: Three-dimensional MMU embedding of the proton “chair” geometry inside the lower base triangle of the tetrahedral ur-cell. Each colored tetrahedral sector represents one of the three principal stiffness directions (w_2, w_3, w_4), while the shared edges form the hexagonal chair ring that defines the proton’s internal torsional backbone. This 3D embedding makes visible how chair spin, radial stiffness, and quark-channel attachment arise from the same geometric ur-cell configuration.

Purpose of this work

In this paper we construct an explicit chair–quark mass–spring model based entirely on MMU stiffness rules evaluated at the proton radius. Our goals are:

1. to derive the uud charge pattern from a geometric eigenvector of the quark ring;
2. to demonstrate the clean separation between chair-dominated (proton shell) modes and quark-dominated inner-ring modes in the eigenfrequency spectrum;
3. to compute geometric torsional spin weights for the three quarks;
4. and to obtain a robust proton g -factor from the universal MMU stiffness ratio k_3/k_2 , without tuned parameters.

The resulting expression,

$$g_p^{\text{MMU}} = \frac{2}{3} \sqrt{\frac{k_3}{k_2}} = \frac{2}{3} \sqrt{\frac{1}{2\alpha}},$$

agrees with experiment at the one-percent level and emerges here as a purely geometric property of torsional elasticity rather than as a phenomenological fit.

2 MMU Stiffnesses and Chair–Quark Model

The MMU stiffness rules for electric and torsional axes are

$$k_2(a) = \frac{2\alpha\hbar c}{a^3}, \quad k_3(a) = \frac{\hbar c}{a^3}, \quad (1)$$

so that

$$\frac{k_3}{k_2} = \frac{1}{2\alpha}, \quad \sqrt{\frac{k_3}{k_2}} = \sqrt{\frac{1}{2\alpha}}. \quad (2)$$

Evaluating at the proton charge radius $a_p \approx 0.84 \text{ fm}$ fixes the absolute stiffness scale, but the ratio k_3/k_2 is independent of a_p and purely geometric.

We consider the following degrees of freedom:

- six radial chair coordinates forming a hexagonal ring,
- one global chair w_3 spin coordinate,
- three quark channels, each with (w_2, w_3, w_4) .

This yields $N = 16$ coordinates in total.

The chair ring uses the standard hexagonal Laplacian scaled by $k_2(a_p) + k_3(a_p)$. The global chair spin coordinate carries a torsional stiffness of order $6k_3(a_p)$. Quark channels are modelled as effectively shorter MMU edges with $a_q \approx 0.5a_p$, such that their internal stiffnesses increase via the a^{-3} law. Chair–quark couplings are chosen small and diagonal in (w_2, w_3, w_4) to represent weak elastic contact between shell and inner ring.

All coupling strengths, cross–spring coefficients, and effective masses enter only as *dimensionless* parameters. Crucially, no parameter is tuned to reproduce the proton g -factor.

3 Geometric Origin of the uud Charges

On the quark ring, three quark nodes are arranged on a hexagon. The antisymmetric ring mode is

$$v = (1, 1, -2), \quad (3)$$

orthogonal to the symmetric mode $(1, 1, 1)$ and describing a charge–neutral torsional pattern.

We assume a linear projection from this eigenvector to quark charges:

$$q_i = q_0 + \alpha_q v_i. \quad (4)$$

Enforcing the proton charge $\sum_i q_i = 1$ and symmetry between quarks 1 and 2 yields

$$q_0 = \frac{1}{3}, \quad \alpha_q = \frac{1}{3}. \quad (5)$$

Hence

$$(q_1, q_2, q_3) = \left(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3} \right), \quad (6)$$

which is the standard QCD uud pattern obtained from pure geometry, without invoking QCD field theory.

This pattern controls how torsional w_3 spin on the quark ring contributes to the electromagnetic moment and will enter directly into the definition of mode g -factors.

4 Eigenmode Spectrum of the Chair–Quark System

We assemble a global stiffness matrix K and mass matrix M from:

- a hexagonal chair ring submatrix acting on six radial chair coordinates,
- a torsional stiffness for the global chair w_3 coordinate,
- local (w_2, w_3, w_4) stiffness matrices for each quark channel,
- quark–ring couplings in w_2 and w_3 ,
- chair–quark couplings in (w_2, w_3, w_4) .

Solving the generalized eigenproblem

$$Kv = \lambda Mv, \quad \omega = \sqrt{\lambda}, \quad E = \hbar\omega, \quad (7)$$

we obtain a discrete set of eigenfrequencies and eigenmodes. With the MMU stiffnesses evaluated at the proton radius, a typical spectrum (for moderate couplings) is:

Mode	Energy [MeV]	Chair fraction	Quark fraction
1	0.0	0.40	0.60
2	0.0	0.00	1.00
3	0.0	0.00	1.00
4	18.9	0.40	0.60
5	118.0	0.00	1.00
6	189.9	0.00	1.00
7	189.9	0.00	1.00
8	290.0	1.00	0.00
9	290.0	1.00	0.00
10	407.2	1.00	0.00
11	502.3	1.00	0.00
12	502.3	1.00	0.00
13	580.0	1.00	0.00
14	997.7	0.00	1.00
15	1577.1	0.00	1.00
16	1577.1	0.00	1.00

Three modes are (approximately) zero-energy, corresponding to global symmetries and gauge-like motion. Nonzero modes fall into two groups:

- **chair-dominated:** modes with $E \sim 100\text{--}600$ MeV and chair fraction close to 1, representing internal shell dynamics and chair spin,
- **quark-dominated:** modes with $E \sim 1.0\text{--}1.6$ GeV and quark fraction close to 1, representing torsional and electric motion on the quark ring.

This separation reflects the MMU picture in which quark channels are stiffer (via shorter edge length) and therefore oscillate at higher energies than the chair shell.

5 Torsional Spin Modes and Quark Spin Weights

The torsional coordinate w_3 is identified with internal MMU spin. For a given eigenmode of the coupled system, let $w_{3,i}$ denote the quark-ring w_3 amplitude of quark i . The spin weight carried by quark i in that mode is

$$S_i = \frac{|w_{3,i}|^2}{\sum_j |w_{3,j}|^2}, \quad \sum_i S_i = 1. \quad (8)$$

The effective *mode* g -factor for this torsional pattern is

$$g_{\text{mode}} = \sum_i q_i S_i, \quad (9)$$

with q_i from the uud pattern (6). This assigns more magnetic weight to positively charged quarks that carry a larger share of the torsional vibration.

In the numerical eigenmode analysis, several high-energy modes ($E \sim 1$ GeV and 1.6 GeV) are both quark-dominated and w_3 -dominated. Typical examples are:

- A mode near $E \approx 1.0$ GeV with nearly symmetric torsion on all quarks, yielding $S \approx (1/3, 1/3, 1/3)$ and $g_{\text{mode}} \approx 1/3$.
- Two modes near $E \approx 1.6$ GeV with torsion concentrated on different quarks, leading to more asymmetric spin weights and g_{mode} values between ~ 0.05 and ~ 0.6 .

These w_3 -dominated modes explore the space of order-one distributions over the three quarks. As we will see, their detailed pattern is less important than the average value emerging from the weighting procedure in the next section.

6 Global Proton g -Factor from MMU Torsion

To obtain a single effective proton g -factor from all modes, we define a weighted average over torsional eigenmodes:

$$g_p^{\text{eff}} = \frac{\sum_{\text{modes}} w_{\text{mode}} g_{\text{mode}}}{\sum_{\text{modes}} w_{\text{mode}}}, \quad (10)$$

where g_{mode} is given by Eq. (9) and w_{mode} encodes how strongly a mode contributes to the proton's magnetic response.

In the present implementation, the weight combines:

- the total quark w_3 amplitude of the mode,
- the global chair–spin amplitude,
- an effective lever arm interpolating between quark radius a_q and proton radius a_p ,
- and an $1/E$ factor suppressing very high–energy modes.

For the class of chair–quark models studied here, the numerical average yields

$$g_p^{\text{eff}} \approx 0.667 \approx \frac{2}{3} \quad (11)$$

for all tested parameter sets. This value is essentially fixed by the uud charges and the symmetry properties of the w_3 –dominated quark modes.

The *physical* proton g –factor in MMU units results from the effective ratio of torsional to electric stiffness. Since the MMU model relates w_3 (spin) and w_2 (electric) oscillations through their stiffnesses, the magnetic response naturally contains the factor

$$\sqrt{\frac{k_3}{k_2}} = \sqrt{\frac{1}{2\alpha}}. \quad (12)$$

Thus the MMU prediction is

$$g_p^{\text{MMU}} = g_p^{\text{eff}} \sqrt{\frac{k_3}{k_2}} \approx \frac{2}{3} \sqrt{\frac{1}{2\alpha}}. \quad (13)$$

Evaluating Eq. (13) with the CODATA value of α yields

$$g_p^{\text{MMU}} \approx 5.518, \quad (14)$$

while the experimental value is $g_p^{\text{exp}} \approx 5.5857$. The relative deviation is about 1.2%.

7 Numerical Robustness

To test whether Eq. (13) is an artefact of a specific parameter choice, we performed extensive scans over:

- chair–quark radial couplings $\varepsilon_{\text{rad}} \in [0.01, 0.10]$,
- chair–quark spin couplings $\varepsilon_{\text{spin}} \in [0.01, 0.10]$,
- cross–spring scales $\beta_{\text{scale}} \in [0.5, 2.0]$,
- effective chair spin mass factors $m_{\text{spin}} \in [0.5, 2.0]$,
- quark radius scales $a_q/a_p \in [0.4, 1.0]$,
- effective quark mass factors $m_q/m_p \in [0.5, 2.0]$.

For all combinations, the numerical code returned

$$g_p^{\text{eff}} \approx 0.667, \quad g_p^{\text{phys}} = \sqrt{\frac{k_3}{k_2}} g_p^{\text{eff}} \approx 5.518, \quad (15)$$

with numerical variations only at the level of floating-point noise. The difference

$$\Delta g = g_p^{\text{phys}} - g_p^{\text{MMU formula}} \quad (16)$$

remained consistent with zero within numerical accuracy.

This suggests that Eq. (13) is not a tuned feature of a particular mechanical model, but a structural consequence of:

1. the uud charge pattern from the antisymmetric ring mode,
2. the MMU stiffness ratio $k_3/k_2 = 1/(2\alpha)$.

8 Discussion and Outlook

We have constructed a unified MMU proton-quark model that:

1. derives the uud charge pattern from a geometric ring eigenvector,
2. exhibits a clear separation between chair and quark modes in the eigenfrequency spectrum,
3. distributes torsional spin over quarks via w_3 -dominated modes,
4. and predicts a proton g -factor

$$g_p^{\text{MMU}} = \frac{2}{3} \sqrt{\frac{1}{2\alpha}}$$

in agreement with experiment at the percent level.

The factor $2/3$ originates from the geometric distribution of w_3 spin over quarks with charges $(2/3, 2/3, -1/3)$; the factor $\sqrt{1/(2\alpha)}$ comes from the universal MMU stiffness ratio between torsional and electric axes. In this sense, the proton magnetic moment becomes a derived quantity, fixed by geometry and the fine-structure constant.

Future work should address:

- corrections from volumetric w_4 and electro-gravity coupling,
- relativistic (Dirac) projections of the internal MMU dynamics,
- extension to neutron-proton mass and g -factor differences,
- application to baryon families and $(g - 2)$ patterns.

If successful, this would support the view that baryon magnetic moments, usually treated as effective parameters in QCD, can be understood as emergent quantities from a microscopic geometric elastic lattice.

References

- [1] J. Wollbold. *A Geometric–Fractal Tetrahedral Model of Matter and Space: From Atomic Bonds to Cosmic Structures*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/HYAX5>.
- [2] J. Wollbold. *From Geometry to Spectra: Predicting Atomic Masses and Ionization Energies from Space Elasticity in the MMU Framework*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/JC2S4>.
- [3] J. Wollbold. *Geometric Origin of the 21 cm Line and Hyperfine Spin-Flip Structure in the Methane Metauniverse*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/JKV6H>.
- [4] J. Wollbold. *Unified Determination of the Cross-Coupling Constants k_{23} , k_{24} and k_{34} in the MMU Tetrahedral Model from Larmor, Zeeman and Lamb Resonances*. OSF Preprint (2025). DOI: <httpsdoi.org/10.17605/OSF.IO/48EQX>.
- [5] J. Wollbold. *Determination of the UR–Tetrahedron in the Methane Metauniverse (MMU): From Elastic Constants to Lepton Scaling*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/NRW3S>.
- [6] J. Wollbold. *Dissipative Lagrange Field Equation of the Methane Metauniverse (MMU): Geometric Origin of Thermal Damping and Spatial Healing*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/VUTNX>.
- [7] J. Wollbold. *The Methane Metauniverse (MMU): A Case Study in AI-Assisted Theoretical Physics and Human–Machine Scientific Creativity*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/SCXPQ>.
- [8] J. Wollbold. *Geometric Quantization of the Internal Edge Length in the MMU Model*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/NUPXY>.
- [9] J. Wollbold. *MMU 2.0: A Unified Geometric Model of Internal Spin, Torsion–Volume Coupling, and the g –Factor in Dual Tetrahedral Spacetime*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/N754E>.
- [10] J. Wollbold. *Geometric Derivation of the Planck Scale in the Methane Metauniverse (MMU)*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/42K9S>.
- [11] J. Wollbold. *Discovery of an Electro–Gravity Fixed Point Beyond the Planck Scale in the MMU Tetrahedral Model*, OSF Preprint (2025), <https://doi.org/10.17605/OSF.IO/ACQNH>.
- [12] J. Wollbold. *From Length–Dependent to Mass–Dependent Stiffness in the MMU: $k_i(a) \rightarrow k_i(m)$ and Leptonic Balmer Predictions*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/V43AY>.
- [13] J. Wollbold. *Projection Principle for Quantum Orbitals: Angular Momentum, Spin, and the Pauli Exclusion Rule as Elastic Torsion in the MMU Tetrahedral Model*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/863MV>.

- [14] J. Wollbold. *The Dirac Equation as the External Projection of the MMU Ur-Cell: A Geometric Origin of Spinors, Antiparticles, and Relativistic Dynamics*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/HG9FN>.
- [15] J. Wollbold. *Geometric Quantization of Space and the Origin of the Rydberg Constant in the MMU*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/RJXE5>.
- [16] J. Wollbold. *Electromagnetic Constants from Tetrahedral Geometry: A Geometric Derivation of ϵ_0 , μ_0 , Z_0 and c in the Methane Metauniverse (MMU)*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/AB4F2>.
- [17] J. Wollbold. *Lepton Masses, Koide Geometry, and the $g - 2$ Family Pattern from Tetrahedral Torsion in the Methane Metauniverse (MMU)*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/GU9RQ>.
- [18] J. Wollbold. *Proton Chair Eigenmodes in the Methane Metauniverse (MMU): From Internal Rotor Frequencies to Hydrogenic Space Response*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/BKDCP>.
- [19] J. Wollbold. *Gravitationally Induced Shifts in Atomic Spectra: A Testable Prediction of Tetrahedral Spacetime Geometry*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/U8B4E>.
- [20] J. Wollbold. *Microscopic Spacetime Kinematics and the Emergence of Einstein Gravity from the Tetrahedral Geometry of the MMU*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/W96H3>.
- [21] J. Wollbold. *MMU 3.0: A Minimal Geometric–Elastic Spacetime Model and the Emergence of Quantum, Relativistic, and Electromagnetic Structure*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/BDTA6>.
- [22] J. Wollbold. *Why Matter Requires Antimatter: A Stability Mechanism in the Dual–Tetrahedral MMU Geometry*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/JYRHK>.
- [23] J. Wollbold. *A Unified Geometric Prediction of Balmer, Zeeman and Lamb Isotope Shifts*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/MZRQ3>.
- [24] J. Wollbold. *Neutron–Proton Mass Gap as an Electro–Gravity Phase Transition in the MMU*. OSF Preprint (2025). DOI: <https://doi.org/10.17605/OSF.IO/SD58Q>.

Listing 1: Python code used to compute the MMU chair-quark spectrum and the proton g -factor.

```
import numpy as np
import matplotlib.pyplot as plt

# =====
# 1. Physikalische Konstanten und MMU-Steifigkeiten k2(a), k3(a), k4(a)
# =====

alpha = 1 / 137.035999084      # Feinstrukturkonstante
hbar   = 1.054571817e-34      # J s
c      = 2.99792458e8         # m/s
eV     = 1.602176634e-19      # J
MeV    = 1e6 * eV            # J
m_p    = 1.67262192369e-27    # kg (Protonmasse)
G      = 6.67430e-11          # m^3 kg^-1 s^-2

hbar_c = hbar * c              # J m

def k2_MMU(a):
    """MMU electric stiffness k2(a) = 2 alpha hbar c / a^3"""
    return 2.0 * alpha * hbar_c / a**3

def k3_MMU(a):
    """MMU torsional stiffness k3(a) = hbar c / a^3"""
    return 1.0 * hbar_c / a**3

def k4_MMU(a, m):
    """MMU volumetric stiffness k4(a) = 2 G m^2 / a^3"""
    return 2.0 * G * m**2 / a**3

# Proton-"chair"-Radius
a_p = 0.84e-15 # m

# Proton-Steifigkeiten direkt aus MMU-Formeln
k2_p = k2_MMU(a_p)
k3_p = k3_MMU(a_p)
k_ring_p = k2_p + k3_p

print("=== Proton chair MMU stiffnesses (from MMU k2,k3 formulas) ===")
print(f"k2_p      = {k2_p:.3e} J/m^2   (Coulomb / w2)")
print(f"k3_p      = {k3_p:.3e} J/m^2   (torsion / w3)")
print(f"k_ring_p   = {k_ring_p:.3e} J/m^2   (effective ring stiffness)")
print()

# Massen (Basiswerte, m_spin_chair wird unten skaliert)
m_eff_chair_base = m_p / 6.0      # radiale Chair-Segmentmasse
m_spin_chair_base = m_p / 2.0     # effektive Chair-Spin-Masse

# =====
# 2. DOF-Z hlung
# =====

N_CHAIR_RAD = 6      # 6 radiale Chair-DOFs
N_CHAIR_SPIN = 1     # 1 globaler Chair-w3-DOF
N_CHAIR = N_CHAIR_RAD + N_CHAIR_SPIN # = 7
N_QUARK = 9          # 3 Quarks x (w2,w3,w4)
DIM = N_CHAIR + N_QUARK # = 16
```

```

# Indizes
IDX_CHAIR_RAD = np.arange(0, N_CHAIR_RAD) # 0..5
IDX_CHAIR_SPIN = N_CHAIR_RAD # 6
FIRST_QUARK = N_CHAIR # 7
IDX_QUARK = np.arange(FIRST_QUARK, FIRST_QUARK + N_QUARK)

# =====
# 3. Globale Cross-Federn (werden im Scan skaliert)
# =====

BETA23 = 0.2
BETA24 = 0.15
BETA34 = 0.1

# =====
# 4. Quark-Ladungen aus Ring-Eigenvektor (1,1,-2)
# =====

def compute_quark_charges():
    v = np.array([1.0, 1.0, -2.0])
    q0 = 1.0 / 3.0
    alpha_q = 1.0 / 3.0
    return q0 + alpha_q * v # [2/3, 2/3, -1/3]

# =====
# 5. Lokale 3x3-MMU-Steifigkeit (w2,w3,w4) mit globalen Betas
# =====

def local_K(k2, k3, k4):
    """Use global BETA23/24/34 so we can vary them in scans."""
    global BETA23, BETA24, BETA34
    k23 = BETA23 * np.sqrt(k2 * k3)
    k24 = BETA24 * np.sqrt(k2 * k4)
    k34 = BETA34 * np.sqrt(k3 * k4)
    return np.array([
        [k2, k23, k24],
        [k23, k3, k34],
        [k24, k34, k4 ],
    ], dtype=float)

def ring_matrix_axis(k):
    return k * np.array([
        [ 2., -1., -1.],
        [-1.,  2., -1.],
        [-1., -1.,  2.],
    ], dtype=float)

# =====
# 6. Aufbau von K und M f r das Gesamtsystem (16x16)
# =====

def build_K_M(epsilon_rad=0.05, epsilon_spin=0.05, m_spin_chair_factor=1.0)
:
    global m_eff_chair_base, m_spin_chair_base

    K_tot = np.zeros((DIM, DIM), dtype=float)
    M_tot = np.zeros((DIM, DIM), dtype=float)

```

```

# aktuelle Massen
m_eff_chair = m_eff_chair_base
m_spin_chair = m_spin_chair_base * m_spin_chair_factor

# ----- Chair-Ring: 6 radiale DOFs (0..5) -----
K_dimless_chair = np.array([
    [ 2, -1,  0,  0,  0, -1],
    [-1,  2, -1,  0,  0,  0],
    [ 0, -1,  2, -1,  0,  0],
    [ 0,  0, -1,  2, -1,  0],
    [ 0,  0,  0, -1,  2, -1],
    [-1,  0,  0,  0, -1,  2]
], dtype=float)

K_chair_rad = k_ring_p * K_dimless_chair
M_chair_rad = np.eye(N_CHAIR_RAD) * m_eff_chair

K_tot[np.ix_(IDX_CHAIR_RAD, IDX_CHAIR_RAD)] = K_chair_rad
M_tot[np.ix_(IDX_CHAIR_RAD, IDX_CHAIR_RAD)] = M_chair_rad

# ----- Chair-Spin-DOF (globaler w3) bei Index 6
# -----
i_spin = IDX_CHAIR_SPIN
# Torsions-Steifigkeit des Rings ~ 6 * k3_p
K_tot[i_spin, i_spin] = 6.0 * k3_p
M_tot[i_spin, i_spin] = m_spin_chair

# ----- Quark-Block (9 DOFs) ab Index FIRST_QUARK
# -----
a_q = a_p / 2.0          # halbe L nge
m_q = m_p / 3.0          # grob 3 Quarks im Proton

k2_q = k2_MMU(a_q)
k3_q = k3_MMU(a_q)
k4_q = k4_MMU(a_q, m_q)

K_loc_q = local_K(k2_q, k3_q, k4_q)

for q in range(3):
    base = FIRST_QUARK + 3*q
    K_tot[base:base+3, base:base+3] += K_loc_q

R2_q = ring_matrix_axis(k2_q * 0.5)
R3_q = ring_matrix_axis(k3_q * 0.5)

for a in range(3):
    for b in range(3):
        ci = FIRST_QUARK + 3*a
        cj = FIRST_QUARK + 3*b
        # w2
        K_tot[ci + 0, cj + 0] += R2_q[a, b]
        # w3
        K_tot[ci + 1, cj + 1] += R3_q[a, b]

# Quark-Massen
m_quark_total = m_p / 3.0
m_quark_dof = m_quark_total / 3.0

```

```

for q in range(3):
    base = FIRST_QUARK + 3*q
    for comp in range(3):
        M_tot[base+comp, base+comp] = m_quark_dof

# ----- C h a i r Quark -Kopplung (radial) -----
gamma2_base = 0.1 * k2_p
gamma3_base = 0.1 * k3_p
gamma4_base = 0.1 * k3_p

gamma2 = epsilon_rad * gamma2_base
gamma3 = epsilon_rad * gamma3_base
gamma4 = epsilon_rad * gamma4_base

chair_nodes = [0, 2, 4] # radiale Chair-DOFs
for q, node in enumerate(chair_nodes):
    base_q = FIRST_QUARK + 3*q
    for comp, g in enumerate([gamma2, gamma3, gamma4]):
        i = node
        j = base_q + comp
        K_tot[i, j] += g
        K_tot[j, i] += g

# ----- C h a i r S p i n Quark -w3-Kopplung -----
gamma3_spin_base = 0.1 * k3_p
gamma3_spin = epsilon_spin * gamma3_spin_base

for q in range(3):
    base_q = FIRST_QUARK + 3*q
    j_w3 = base_q + 1
    K_tot[i_spin, j_w3] += gamma3_spin
    K_tot[j_w3, i_spin] += gamma3_spin

return K_tot, M_tot, a_q

# =====
# 7. Eigenproblem und Energien
# =====

def solve_modes(K, M):
    m_vec = np.diag(M)
    if np.any(m_vec <= 0):
        raise ValueError("Non-positive masses in diagonal of M.")
    Minv_sqrt = np.diag(1.0 / np.sqrt(m_vec))

    K_eff = Minv_sqrt @ K @ Minv_sqrt
    evals, U = np.linalg.eigh(K_eff)
    evals[evals < 0] = 0.0
    omega = np.sqrt(evals)

    idx = np.argsort(omega)
    omega = omega[idx]
    U = U[:, idx]

    V = Minv_sqrt @ U
    E_J = hbar * omega
    E_MeV = E_J / MeV
    return E_MeV, V

```



```

# =====
# 8. Effektiver Protonen-g-Faktor
# =====

def compute_proton_g_eff(E_MeV, V, q_charges, a_p, a_q, energy_power=1.0):
    """
    Effektiver g_p:
    - nutzt Quark-w3-Spinverteilung (Ladungen)
    - Chair-Spin-DOF verst rkt das Moment ber a_p
    - Gewicht:
        weight_m ~ (Q*w3_quark + C_spin) * a_eff / E^energy_power
    """
    chair_rad_idx = IDX_CHAIR_RAD
    i_spin = IDX_CHAIR_SPIN
    quark_idx = IDX_QUARK

    num = 0.0
    den = 0.0

    for Em, vec in zip(E_MeV, V.T):
        v = vec / np.linalg.norm(vec)

        C_rad = np.sum(v[chair_rad_idx]**2)
        C_spin = v[i_spin]**2
        Q = np.sum(v[quark_idx]**2)

        # Quark w2/w3/w4 + w3-Komponenten je Quark
        w2_q = w3_q = w4_q = 0.0
        q_w3 = []
        for q in range(3):
            base = FIRST_QUARK + 3*q
            w2_q += v[base+0]**2
            w3_q += v[base+1]**2
            w4_q += v[base+2]**2
            q_w3.append(v[base+1])
        q_w3 = np.array(q_w3)

        if (w3_q + C_spin) <= 0 or Em <= 0:
            continue

        if np.sum(q_w3**2) == 0:
            continue
        S = q_w3**2 / np.sum(q_w3**2)
        g_mode = np.sum(q_charges * S)

        # effektiver Hebelarm
        a_eff = (w3_q * a_q + C_spin * a_p) / (w3_q + C_spin)

        # Gewichtung: Quark-Spin + Chair-Spin, skaliert mit Radius /
        # Energie^p
        weight = (Q * w3_q + C_spin) * (a_eff / (Em**energy_power))

        num += weight * g_mode
        den += weight

    q_spin_eff = num / den if den > 0 else 0.0
    g_p_eff = 2.0 * q_spin_eff

```

```

        return g_p_eff

# =====
# 9. Hilfsfunktion: ein Modelllauf mit Parametern
# =====

def run_model(eps_rad, eps_spin, beta_scale, m_spin_factor):
    global BETA23, BETA24, BETA34

    # Betas skalieren
    BETA23 = 0.2 * beta_scale
    BETA24 = 0.15 * beta_scale
    BETA34 = 0.1 * beta_scale

    q_charges = compute_quark_charges()
    K, M, a_q = build_K_M(epsilon_rad=eps_rad,
                          epsilon_spin=eps_spin,
                          m_spin_chair_factor=m_spin_factor)
    E_MeV, V = solve_modes(K, M)

    g_p_eff = compute_proton_g_eff(E_MeV, V, q_charges,
                                   a_p=a_p, a_q=a_q,
                                   energy_power=1.0)

    scale = np.sqrt(k3_p / k2_p)
    g_p_phys = scale * g_p_eff
    return g_p_eff, g_p_phys

# =====
# 10. Optional: einfache Spektrums-Analyse f r einen Referenzlauf
# =====

def analyze_modes(E, V):
    n_modes = len(E)
    chair_idx = np.arange(0, N_CHAIR)
    quark_idx = IDX_QUARK

    chair_frac = np.zeros(n_modes)
    quark_frac = np.zeros(n_modes)
    w2_frac = np.zeros(n_modes)
    w3_frac = np.zeros(n_modes)
    w4_frac = np.zeros(n_modes)

    print("\nMode analysis (chair/quark and w2/w3/w4 fractions):\n")
    for m, (Em, vec) in enumerate(zip(E, V.T), start=1):
        v = vec / np.linalg.norm(vec)

        chair_amp = np.sum(v[chair_idx]**2)
        quark_amp = np.sum(v[quark_idx]**2)

        w2 = w3 = w4 = 0.0
        for q in range(3):
            base = FIRST_QUARK + 3*q
            w2 += v[base + 0]**2
            w3 += v[base + 1]**2
            w4 += v[base + 2]**2

        chair_frac[m-1] = chair_amp

```

```

    quark_frac[m-1] = quark_amp
    w2_frac[m-1] = w2
    w3_frac[m-1] = w3
    w4_frac[m-1] = w4

    print(f"Mode {m:2d}: E = {Em:8.3f} MeV")
    print(f"  chair : {chair_amp:5.3f},  quark : {quark_amp:5.3f}")
    print(f"  w2 fraction: {w2:5.3f},  w3 fraction: {w3:5.3f},  w4
          fraction: {w4:5.3f}\n")

    return chair_frac, quark_frac, w2_frac, w3_frac, w4_frac

# =====
# 11. Main mit Robustheits-Scan
# =====

def main():
    # 1) Referenzlauf mit Ausgabe der Eigenenergien + g_p
    print("=== Reference run (detail) ===")
    q_charges = compute_quark_charges()
    print("Quark charges from ring mode (1,1,-2):", q_charges,
          "    sum =", np.sum(q_charges))

    # Referenz-Parameter
    eps_rad_ref = 0.05
    eps_spin_ref = 0.05
    beta_scale_ref = 1.0
    m_spin_factor_ref = 1.0

    BETA23 = 0.2 * beta_scale_ref
    BETA24 = 0.15 * beta_scale_ref
    BETA34 = 0.1 * beta_scale_ref

    K_ref, M_ref, a_q_ref = build_K_M(epsilon_rad=eps_rad_ref,
                                      epsilon_spin=eps_spin_ref,
                                      m_spin_chair_factor=m_spin_factor_ref
                                      )
    E_ref, V_ref = solve_modes(K_ref, M_ref)

    print("\nEigenenergies (MMU k2,k3,k4, chair+spin+quark):")
    for i, Em in enumerate(E_ref, start=1):
        print(f"Mode {i:2d}: {Em:8.3f} MeV")

    analyze_modes(E_ref, V_ref)

    g_p_eff_ref = compute_proton_g_eff(E_ref, V_ref, q_charges,
                                       a_p=a_p, a_q=a_q_ref,
                                       energy_power=1.0)

    scale = np.sqrt(k3_p / k2_p)
    g_p_phys_ref = scale * g_p_eff_ref
    print(f"\nReference: g_p_eff      {g_p_eff_ref:.3f}")
    print(f"Stiffness scale factor sqrt(k3_p/k2_p) = {scale:.3f}")
    print(f"Scaled physical g_p_phys      {g_p_phys_ref:.3f}\n")

    # 2) Grober Scan  ber Parameter
    print("=== Robustness scan over eps_rad, eps_spin, beta_scale,
          m_spin_factor ===")
    for eps_rad in [0.01, 0.05, 0.1]:

```

```

for eps_spin in [0.01, 0.05, 0.1]:
    for beta_scale in [0.5, 1.0, 2.0]:
        for m_spin_factor in [0.5, 1.0, 2.0]:
            g_eff, g_phys = run_model(eps_rad, eps_spin,
                                      beta_scale, m_spin_factor)
            print(f"eps_rad={eps_rad:.2f}, eps_spin={eps_spin:.2f},
                  "
                  f"beta_scale={beta_scale:.1f}, m_spin_factor={
                      m_spin_factor:.1f} "
                  f"-> g_eff={g_eff:.3f}, g_phys={g_phys:.3f}")

if __name__ == "__main__":
    main()

```